STAT 5080: Experimental Design Group Assignment

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1. A life insurance salesman believes that the mean age of people who buy their first life insurance plan is less than 35. To test his belief, he takes a random sample of 15 customers who have just purchased their first life insurance. Their ages are shown, and assume to be normally distributed: 42, 43, 28, 34, 30, 36, 25, 29, 32, 33, 27, 30, 22, 37, and 40. (Please use a statistical software package to solve the problems)

(a) Can we conclude at the 5% significance level that the insurance salesman is correct? (1)

H_0: $\mu \geq 35$ vs. H_a: $\mu < 35$

```
ages <- c(42, 43, 28, 34, 30, 36, 25, 29, 32, 33, 27, 30, 22, 37, 40)

t.test(ages, mu = 35, alternative = "less", conf.level = 0.95)
```

```
##
##  One Sample t-test
##
## data:  ages
## t = -1.5472, df = 14, p-value = 0.07205
## alternative hypothesis: true mean is less than 35
## 95 percent confidence interval:
##      -Inf 35.34128
## sample estimates:
## mean of x
##  32.53333
```

At the 5% significance level, there is insufficient evidence to conclude that the true mean age of customers is less than 35 years. Therefore, the salesman's claim is not supported by the data at $\alpha = 0.05$.

b. Construct a 95% confidence interval of the mean age of people who buy their first life insurance plan. Interpret the confidence interval in the context of the problem. (1)

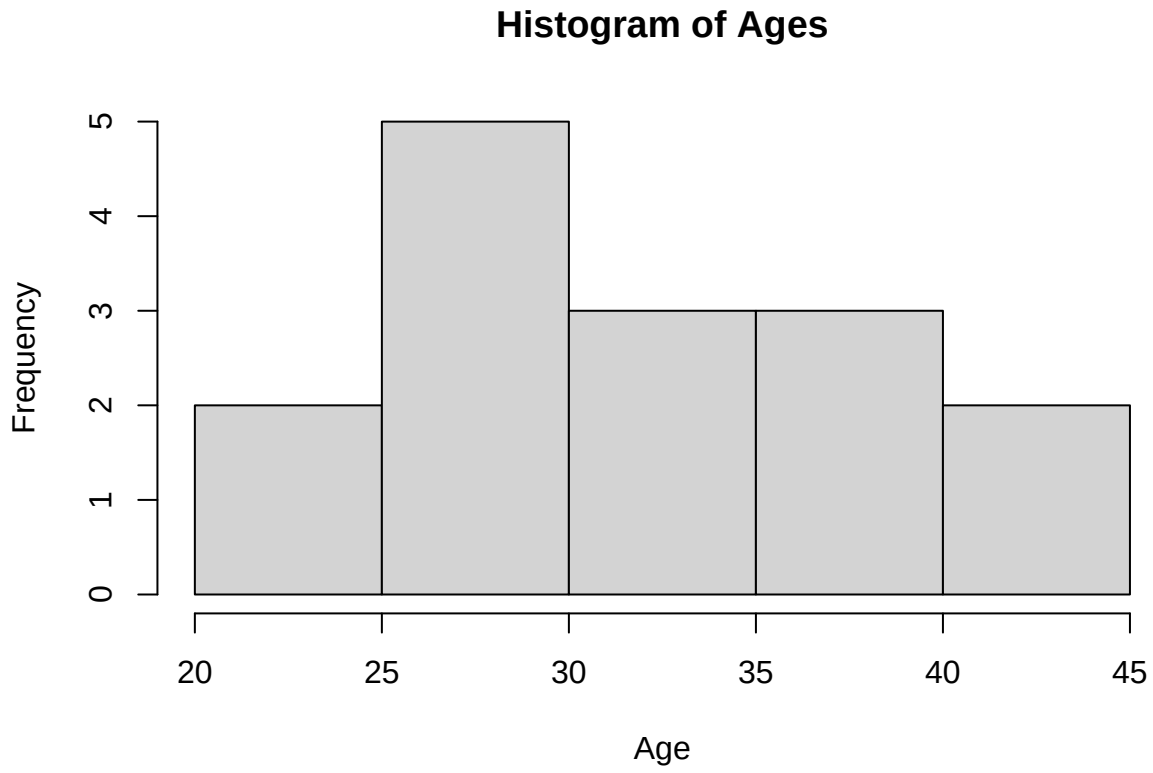
```
t.test(ages, conf.level = 0.95)$conf.int
```

```
## [1] 29.11404 35.95263
## attr(,"conf.level")
## [1] 0.95
```

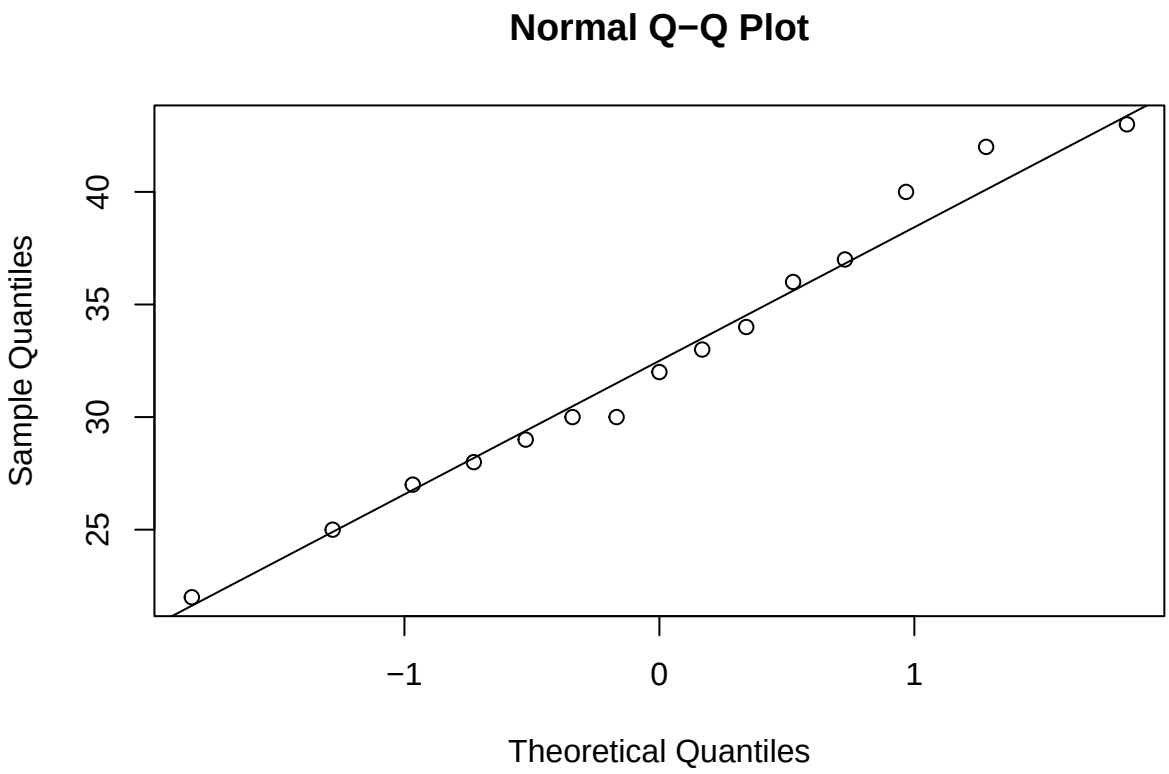
We are 95% confident that the true mean age of people who buy their first life insurance plan is between 29.11 and 35.95 years. Since 35 lies within this interval, the data are consistent with a mean age of 35.

c. Check to ensure that the assumption of the statistical method used in parts (a) and (b) is satisfied. (1)

```
hist(ages, main="Histogram of Ages", xlab="Age")
```




```
qqnorm(ages); qqline(ages)
```




```
# Formal normality test
shapiro.test(ages)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  ages
## W = 0.97387, p-value = 0.9107
```

Because the p-value (0.9107) is much larger than $\alpha = 0.05$, we fail to reject the null hypothesis of normality. In addition, the Q–Q plot shows no serious departures from a straight line, and further ascertained by the histogram.

2. Do government employees take longer coffee breaks on average than do private sector workers? This is a question a management consultant was tasked to find to investigate. In order to investigate the issue, he took a random sample of 100 government employees and another sample of private sector workers. The amount of time (in minutes) spent taking coffee breaks by these individuals were measured and recorded in "Coffee_Break.txt", Coffee_Break.xlsx", and "Coffee_Break.dat". (Please use a statistical software package to solve the problems)

a. Are these two samples independent or dependent? (1)
The two samples are independent. One random sample was taken from government employees and a

separate random sample was taken from private-sector workers. Since the individuals are not paired or matched in any way, the observations in the two groups are independent.

b. Do these data provide significant evidence to support the assertion that government employees take longer coffee breaks on average than private sector workers do? You need to write down the hypotheses. (2)

Let $\mu_{\text{Government}}$ denote the true mean coffee break time (in minutes) for government employees, and let μ_{Private} denote the true mean coffee break time for private-sector employees.

The hypotheses to be tested are

$H_0: \ \mu_{\text{Government}} - \mu_{\text{Private}} = 0$

$H_A: \ \mu_{\text{Government}} - \mu_{\text{Private}} > 0$

```
employees_data <- read.delim("Coffee_Break.txt")

gov <- employees_data$Government
priv <- employees_data$Private

t.test(gov, priv, alternative = "greater", var.equal = FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  gov and priv
## t = 10.503, df = 173.68, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  6.113924      Inf
## sample estimates:
## mean of x mean of y
##  28.04188  20.78545
```

We reject H_0 since the p-value (2.2×10^{-16}) is less than $\alpha = 0.05$. There is overwhelming statistical evidence that the mean coffee break time for government employees is greater than that for private-sector employees.

c. Construct a 95% confidence interval of the difference in the average time spent on coffee breaks between government employees and private sector workers. Interpret the confidence interval in the context of the problem. (2)

```
# 95% CI for mu_Government - mu_Private (Welch)
t.test(gov, priv, conf.level = 0.95, var.equal = FALSE)$conf.int
```

```
## [1] 5.892809 8.620059
## attr(,"conf.level")
## [1] 0.95
```

We are 95% confident that the true difference in mean coffee break times, $\mu_{\text{Government}} - \mu_{\text{Private}}$, lies between 5.89 and 8.62 minutes. This indicates that government employees spend, on average, between about 6 and 9 minutes longer on coffee breaks than private-sector workers.

d. What are the required conditions under which the statistical methods performed in part (b) and part (c) are valid? Please list these conditions and assess the validity of these conditions. Note that you don't necessarily have to perform any formal test. (1)

The conditions are:
i. The populations are independent.
ii. The two samples are assumed to be drawn from a population that is normally distributed.

3. To determine the effect of advertising on the internet, Google took a sample of 400 retails stores that did not advertise on the internet last year but did so this year. The annual sales (in thousands of dollars) in both years for each store were recorded and stored in "Internet Advertise.txt", "Internet Advertise.xlsx", and "Advertise.dat". (Please use a statistical software package to solve the problems)

a. Are these two samples independent or dependent? (1)
The two samples are dependent (paired) because annual sales were measured for the same 400 retail stores in two different years. Each store's sales before advertising are paired with its sales after advertising.

For each store i, define the paired difference $d_i = (\text{Sales}_{i,\text{After Internet Ads}} - \text{Sales}_{i,\text{Before Internet Ads}})$, where sales are measured in thousands of dollars.

Let $\mu_d = E(d_i)$ denote the population mean change in annual sales (After advertising minus Before advertising).

We test whether internet advertising increases mean sales on average:

$H_0 : \ \mu_d = 0 \qquad \text{vs.} \qquad H_A : \ \mu_d > 0.$

```
data_internet <- read.delim("INTERNET ADVERTISE.txt")


this_year <- data_internet$`This.Year`
last_year <- data_internet$`Last.Year`


diff_sales <- this_year - last_year

t.test(this_year, last_year,
       alternative = "greater",
       paired = TRUE,
       conf.level = 0.95)
```

```
##
##  Paired t-test
##
## data:  this_year and last_year
## t = 6.4937, df = 399, p-value = 1.247e-10
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  16.832    Inf
## sample estimates:
## mean difference
##        22.55969
```

At the $\alpha = 0.05$ significance level, we reject H_0 since the p-value is 1.247×10^{-10}. There is strong statistical evidence that advertising on the internet increases average annual sales.

c. Construct a 95% confidence interval of the difference of average sales between the two years. Interpret the confidence interval in the context of the problem. (2)

```
t.test(this_year, last_year,paired = TRUE,conf.level = 0.95)$conf.int
```

```
## [1] 15.72986 29.38951
## attr(,"conf.level")
## [1] 0.95
```

We are 95% confident that the true mean increase in annual sales after advertising on the internet lies between \$15.73 thousand and \$29.39 thousand per store. This interval indicates that internet advertising leads to a positive increase in average sales.

d. What are the required conditions under which the statistical methods performed in part (a) and part (b) are valid? Please list these conditions and assess the validity of these conditions. Note that you don't necessarily have to perform any formal test. (1)

i) Populations are dependent.

ii) Population of differences is normal.